

# Jaehan Kim

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## RESEARCH INTEREST

AI Security (LLM Safety), AI for Cybersecurity

## EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)** March 2022 - February 2026  
Ph.D. (Candidate) in Electrical Engineering, Network and System Security Lab (Advisor: *Seungwon Shin*)  
Dissertation: *Securing Efficient Techniques for Large Language Model Development* Daejeon, South Korea
- **Korea Advanced Institute of Science and Technology (KAIST)** March 2020 - February 2022  
M.S. in Electrical Engineering, Network and System Security Lab (Advisor: *Seungwon Shin*)  
Dissertation: *An Empirical Analysis of Credential Breaches from Vulnerable Websites* Daejeon, South Korea
- **Korea Advanced Institute of Science and Technology (KAIST)** March 2016 - February 2020  
B.S. in Electrical Engineering, Minor in Computer Science Daejeon, South Korea

## PUBLICATIONS [C]: CONFERENCE, [J]: JOURNAL, [U]: UNDER REVIEW

- [C] J. Kim, M. Song, S. Shin, S. Son. **SafeMoE: Safe Fine-Tuning for MoE LLMs by Aligning Harmful Input Routing**. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*
- [J] J. Kim, M. Song, M. Seo, Y. Jin, S. Shin, J. Kim. **PassREfinder-FL: Privacy-Preserving Credential Stuffing Risk Prediction via Graph-Based Federated Learning for Representing Password Reuse between Websites**. *Elsevier Expert Systems with Applications (ESWA)*
- [C] J. Kim, S.H. Na, M. Song, S. Shin, S. Son. **MoEvil: Poisoning Experts to Compromise the Safety of Mixture-of-Experts LLMs**. *Proceedings of the 41th Annual Computer Security Applications Conference (ACSAC 2025)* (**Distinguished Paper Award**)
- [C] J. Kim, M. Song, S.H. Na, S. Shin. **Obliviate: Neutralizing Task-Agnostic Backdoors within the Parameter-Efficient Fine-Tuning Paradigm**. *The 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NACCL 2025 Findings)*
- [C] M. Song, H. Kim, J. Kim, Y. Jin, S. Shin. **Claim-Guided Textual Backdoor Attack for Practical Applications**. *The 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NACCL 2025 Findings)*
- [C] M. Song, H. Kim, J. Kim, S. Shin, S. Son. **Refusal Is Not an Option: Unlearning Safety Alignment of Large Language Models**. *34th USENIX Security Symposium (USENIX Security 2025)*
- [C] M. Song, E. Jang, J. Kim, S. Shin. **Covering Cracks in Content Moderation: Delexicalized Distant Supervision for Illicit Drug Jargon Detection**. *31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2025)*
- [C] M. Seo, M. You, J. Kim, T. Park, S. Shin, J. Kim. **MUFFLER: Secure Tor Traffic Obfuscation with Dynamic Connection Shuffling and Splitting**. *2025 IEEE International Conference on Computer Communications (INFOCOM 2025)*
- [C] G. Park, J. Kim, J. Choi, J. Kim. **CryptoGuard: Lightweight Hybrid Detection and Prevention of Host-Based Cryptojackers**. *Proceedings of the 19th ACM Asia Conference on Computer and Communications Security (ASIACCS 2025)*
- [C] S. Kim, S.H. Na, J. Kim, S. Shin, H. Choi. **AVXProbe: Enhancing Website Fingerprinting with Side-Channel Assisted Kernel-Level Traces**. *Proceedings of the 19th ACM Asia Conference on Computer and Communications Security (ASIACCS 2025)*
- [C] J. Kim, M. Song, M. Seo, Y. Jin, S. Shin. **PassREfinder: Credential Stuffing Risk Prediction by Representing Password Reuse between Websites on a Graph**. *2024 IEEE Symposium on Security and Privacy (S&P 2024)*
- [C] M. You, J. Nam, H. Seo, M. Seo, J. Kim, D. Choi, S. Shin. **HardWhale: A Hardware-Isolated Network Security Enforcement System for Cloud Environments**. *2024 IEEE 44th International Conference on Distributed Computing Systems (ICDCS 2024)*
- [J] M. You, M. Seo, J. Kim, S. Shin, J. Nam. **Hyperion: Hardware-Based High-Performance and Secure System for Container Networks**. *IEEE Transactions on Cloud Computing 2024 (TCC 2024)*

- [J] M. Seo\*, J. Kim\*, M. You, S. Shin, J. Kim. **gShock: A GNN-based Fingerprinting System for Permissioned Blockchain Networks over Encrypted Channels**. *IEEE Access* 2024 \*equally contributed
- [J] S. Lee\*, J. Kim\*, M. Seo, S.H. Na, S. Shin, J. Kim. **CENSor: Detecting Illicit Bitcoin Operation via GCN-Based Hyperedge Classification**. *IEEE Access* 2024 \*equally contributed
- [C] M. Seo, J. Kim, E. Marin, M. You, T. Park, S. Lee, S. Shin, J. Kim. **Heimdallr: Fingerprinting SD-WAN Control-Plane Architecture via Encrypted Control Traffic**. *Proceedings of the 38th Annual Computer Security Applications Conference (ACSAC 2022)*
- [J] M. You, Y. Kim, J. Kim, M. Seo, S. Son, S. Shin, S. Lee. **FuzzDocs: An Automated Security Evaluation Framework for IoT**. *IEEE Access* 2022
- [J] M. You, J. Kim, S. Shin. **Revisiting Security Landscape of Docker Hub Container**. *The Journal of Korean Institute of Communications and Information Sciences* 2022
- [J] J. Choi\*, J. Kim\*, M. Song, H. Kim, N. Park, M. Seo, Y. Jin, S. Shin. **A Large-Scale Bitcoin Abuse Measurement and Clustering Analysis Utilizing Public Reports**. *IEICE Transactions on Information and Systems* 2022 \*equally contributed
- [U] J. Kim, M. Song, M. Kang, H. Kim, S. Shin, S. Son **Subgraph Reconstruction Attacks on Graph RAG Deployments with Practical Defenses**. *arXiv* 2026
- [U] J. Kim, M. Seo, M. Song, S. Shin, J. Kim. **To Make Each Account Count: Exploring Credential Data Breach Threats through Victim-driven Analysis**. *Submitted to Elsevier Computers & Security*
- [U] S. Song, D. Lee, J. Kim, J. Choi, J. Kim. **Assessing Hallucination in Large Language Models for Cyber Threat Intelligence: A First Measurement Study**. *Submitted to Elsevier Engineering Applications of Artificial Intelligence (EAAI)*

## EXPERIENCE

- **S2W**  Dec 2020 - Feb 2021  
 Research Intern @ NLP Team South Korea
  - **Cyber Security Event Detection System**: Proposed a deep learning-based system for extracting security-related events from real-time, unstructured, and noisy data across platforms such as SNS, blogs, and the dark web.
- **SK hynix**  Dec 2018 - Feb 2019  
 Software Engineer Intern @ NAND/Solution PE Team South Korea
  - **Storage Plan for Edge Computing Systems**: Developed a future plan for adopting appropriate storage interfaces based on the workload demands of emerging technologies such as AI.
- **KAIST NSS Lab** Jun 2018 - Dec 2019  
 Undergraduate Research Intern South Korea
  - Developed a CLI extension for security assessment framework for software-defined networks.

## AWARDS

- **Distinguished Paper Award** 2025  
*Annual Computer Security Applications Conference (ACSAC)*
  - MoEvil: Poisoning Expert to Compromise the Safety of Mixture-of-Experts LLMs
- **4th Prize, 2025 Cybersecurity Paper Competition** 2025  
*Korean Association of Cybersecurity Studies (KACS)*
  - Poisoning Expert to Compromise the Safety of Mixture-of-Experts LLMs
- **4th Prize, 2024 Cybersecurity Paper Competition** 2024  
*Korean Association of Cybersecurity Studies (KACS)*
  - LLM Backdoor Defense within the Parameter-Efficient Fine-Tuning Paradigm
- **2nd Prize, 2023 Cybersecurity Paper Competition** 2023  
*Korean Association of Cybersecurity Studies (KACS)*
  - Graph-based Deep Learning Framework for Credential Stuffing Risk Prediction
- **4th Prize, 2023 Cybersecurity Paper Competition** 2023  
*Korean Association of Cybersecurity Studies (KACS)*
  - Delexicalized Distant Supervision for Illicit Drug Jargon Detection
- **4th Prize, 2023 Cybersecurity Paper Competition** 2023  
*Korean Association of Cybersecurity Studies (KACS)*
  - Understanding the Occurrence and Impact of Credential Data Breach

- **4th Prize, 29th Samsung Humantech Paper Awards**

2023

*Samsung Electronics*

- Heimdallr: Fingerprinting SD-WAN Control-Plane Architecture via Encrypted Control Traffic

- **4th Prize, 2021 Cybersecurity Paper Competition**

2021

*Korea Institute of Information Security and Cryptology (KIISC)*

- Fingerprinting Distributed SD-WAN Control-Plane Architecture via Encrypted Control Traffic